

1-bit Full adder

A	B	Cin	D	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

D

A \ BCin	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$D = A \oplus B \oplus C_{in}$$

Cout

A \ BCin	00	01	11	10
0	0	0	1	0
1	0	1	1	1

Cin

$$C_{out} = BC_{in} + AB + AC_{in}$$